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Robust CurveMoE: Multi-Norm Adversarial Defense for Mixture-of-Experts Models via Mode Connectivity

Low-Loss Bézier Paths Through MoE Weight Space Combine Per-Norm Robust Endpoints Into a Single Multi-Norm Robust Model

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Mixture-of-Experts models are increasingly deployed at scale, but their robustness to adversarial perturbations across multiple ℓ_p threat models simultaneously has been understudied. This paper proposes Robust CurveMoE: train separate MoE models robust under each threat norm, find low-loss Bézier curve paths connecting them through weight space, and sample a final model from the connecting path. The resulting model achieves multi-norm adversarial robustness without ensemble inference overhead, with clean accuracy dropping only 1.2% relative to the endpoint models.

AT A GLANCE

Problem: MoE models trained for robustness under one ℓ_p norm are vulnerable to other norms; union adversarial training is expensive and hurts accuracy.

Why it matters: Production systems face diverse adversarial attacks; a single model must defend across all without multiplying inference cost.

Method: Find Bézier curve paths through MoE weight space connecting per-norm robust endpoints; sample from the path at the point of best multi-norm trade-off.

Result: Robust accuracy [86.2%, 71.4%, 68.3%] under $[\ell_\infty, \ell_2, \ell_1]$ vs. 4.8% clean accuracy drop for union AT; CurveMoE drops only 1.2%.

Evidence: ResNet MoE on CIFAR-100; ablations on path degree and Gumbel-softmax relaxation.

The Big Picture

Adversarial robustness is typically studied under a single threat model: ℓ_∞ -bounded perturbations with radius ϵ_∞ . But real-world adversaries can construct attacks in any ℓ_p geometry, and a model robust to ℓ_∞ attacks may be easily fooled by ℓ_2 or ℓ_1 perturbations.

Multi-norm robustness — simultaneous robustness to ℓ_∞ , ℓ_2 , and ℓ_1 threats — has been addressed for dense net-

works by union adversarial training or ensembles, but both approaches fail for Mixture-of-Experts models:

Union AT trains on the worst-case perturbation over all three norms simultaneously. This is computationally expensive and leads to an accuracy-robustness trade-off that worsens as the number of norms increases.

Ensembles combine predictions from three separately robust models, tripling inference cost. For MoE models with thousands of experts, ensemble inference is prohibitive.

Mode connectivity offers a third path: connect separately trained robust models through low-loss paths in weight space and sample a model from the path. For dense networks this has been demonstrated; for MoE models, the routing-dependent loss landscape poses new challenges.

Why Existing Approaches Fall Short

Single-norm AT: Provides strong robustness under the target norm but typically transfers poorly to orthogonal norms. A model ℓ_∞ -trained at $\epsilon = 8/255$ may have robust accuracy below 40% under ℓ_2 attacks of matched strength.

Union AT for MoE: The multi-norm adversarial objective requires solving $\max_{p \in \{\infty, 2, 1\}} \max_{\|\delta\|_p \leq \epsilon_p} \mathcal{L}(\theta; x + \delta, y)$ at each training step. For MoE models, the inner max interacts with discrete expert routing: the worst-case perturbation may change the routing path, invalidating the gradient from the current routing.

Mode connectivity for dense networks: Garipov et al. (2018) showed that dense network pairs trained independently can be connected by low-loss quadratic Bézier curves. For MoE models, the argmax routing creates discontinuities that break the Garipov construction; straight-line interpolation passes through high-loss regions.

Core Mechanism

The CurveMoE approach has four stages:

1. **Per-norm endpoint training:** Train $\theta_{\ell_\infty}, \theta_{\ell_2}, \theta_{\ell_1}$ using standard adversarial training under each norm respectively.
2. **Gumbel-softmax relaxation:** Replace argmax expert routing with Gumbel-softmax to make the loss differentiable with respect to weight interpolations along the connecting path.
3. **Path optimization:** Find a Bézier curve $\phi(t)$ connecting two endpoint models by minimizing the expected loss along the path under the relaxed routing.
4. **Path sampling:** Evaluate a grid of t values and select t^* that maximizes the minimum robust accuracy across all three norms on a held-out validation set.

For three endpoints, a triangular convex combination is used instead of a pairwise curve, providing simultaneous

connectivity to all three robust models.

Technical Formulation

A degree-2 Bézier curve between θ_A and θ_B with midpoint θ_c is:

$$\phi(t) = (1-t)^2\theta_A + 2t(1-t)\theta_c + t^2\theta_B, \quad t \in [0, 1]$$

The midpoint θ_c is optimized by minimizing:

$$\mathcal{L}_{\text{path}}(\theta_c) = \mathbb{E}_{t \sim U[0,1]} \sum_{p \in \{\infty, 2, 1\}} \alpha_p \mathcal{L}_p^{\text{adv}}(\phi(t))$$

where $\mathcal{L}_p^{\text{adv}}(\theta) = \max_{\|\delta\|_p \leq \epsilon_p} \mathcal{L}(\theta; x + \delta, y)$.

For the three-endpoint triangular path with $t_1 + t_2 \leq 1$:

$$\phi(t_1, t_2) = t_1\theta_{\ell_\infty} + t_2\theta_{\ell_2} + (1-t_1-t_2)\theta_{\ell_1}$$

with three midpoint parameters $\theta_{c_{12}}, \theta_{c_{13}}, \theta_{c_{23}}$ along each edge of the triangle.

The Gumbel-softmax relaxation at temperature τ_g :

$$\hat{g}_i = \frac{\exp((h_i + g_i)/\tau_g)}{\sum_j \exp((h_j + g_j)/\tau_g)}$$

where $g_i \sim \text{Gumbel}(0, 1)$ and h_i are expert logits. During path optimization $\tau_g = 1.0$; reverted to hard routing ($\tau_g \rightarrow 0$) for evaluation.

Learning or Inference Procedure

1. Train θ_{ℓ_p} for each $p \in \{\infty, 2, 1\}$ using PGD – 10 adversarial training under the respective norm.
2. Initialize pairwise Bézier midpoints $\theta_c = (\theta_A + \theta_B)/2$.
3. Optimize θ_c for 50 epochs with Gumbel-softmax routing at $\tau_g = 1.0$, Adam optimizer, learning rate 10^{-3} .
4. Grid-search $t \in \{0, 0.05, \dots, 1.0\}$ on the validation set; select t^* maximizing $\min_p R_p(\phi(t^*))$ where R_p is robust accuracy under norm p .
5. The final model is $\phi(t^*)$ with hard routing; no inference overhead.

What the Guarantee Says

The path-loss objective guarantees that $\hat{V}(x_t)$ (the loss along the optimized path) is uniformly low across $t \in [0, 1]$. Because the path connects models robust under each individual norm, and the path loss is low (implying robustness is maintained along the path), models at intermediate t values inherit multi-norm robustness properties from both endpoints.

This is not a formal robustness certificate but a constructive argument: the path endpoint models have certified robustness under their respective norms, and the path loss bounds how much that robustness degrades along the path.

Experimental Findings

Evaluated on ResNet-56 MoE with 8 experts on CIFAR-100, under PGD-50 attacks with $\ell_\infty \epsilon = 8/255$, $\ell_2 \epsilon = 0.5$, $\ell_1 \epsilon = 12$:

- **CurveMoE robust accuracy:** [ℓ_∞ : 86.2%, ℓ_2 : 71.4%, ℓ_1 : 68.3%]
- **Best single-norm AT (per norm):** [ℓ_∞ : 89.1%, ℓ_2 : 74.2%, ℓ_1 : 71.8%] (different models for each)
- **Union AT:** [ℓ_∞ : 81.4%, ℓ_2 : 68.7%, ℓ_1 : 65.2%] with –4.8% clean accuracy drop
- **CurveMoE clean accuracy:** –1.2% relative to endpoint models
- **Path loss:** Maximum loss along the optimized path is 4% higher than at endpoints; linear interpolation paths peak at 31% higher, confirming the need for degree-2 curves.

Ablations and Interpretation

Path degree: Linear interpolation (degree 1) passes through high-loss regions for MoE models. Degree-2 Bézier curves find a low-loss path for all endpoint pairs tested. Degree 3 provides marginal improvement.

Gumbel temperature: $\tau_g = 1.0$ during path optimization gives smooth gradients; lower temperatures ($\tau_g = 0.1$) approximate hard routing but produce noisy gradients; $\tau_g = 0.5$ is a useful intermediate for larger MoEs.

Number of experts: Path optimization scales well to 16 and 32 experts; beyond 64 experts, path optimization requires more epochs to overcome routing instability.

Transfer to ViT-MoE: The approach transfers to Vision Transformer MoE architectures with token routing; mode connectivity persists, though the midpoint loss is 6% higher than for ResNet MoE, suggesting transformer MoE landscapes are less connected.

Reference

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